

PREDICTED PHYSICS BOARD PAPER 2026

Based on Serial Repetition Analysis of 2023-2025 Sets

MAX MARKS: 70

TIME: 3 HOURS

SECTION A (Multiple Choice)

Q1. Two nuclei have mass numbers in the ratio 27:64. The ratio of their nuclear densities is:

- (a) 3:4 (b) 9:16 (c) 1:1 (d) 27:64

Repetition: Set 55/7/1, 55/7/2, 55/5/2

Q2. When a p-n junction diode is forward biased, the width of the depletion layer:

- (a) remains same (b) decreases (c) increases (d) becomes double

Q3. The slope of the stopping potential versus frequency graph for a photosensitive metal gives:

- (a) h (b) h/e (c) e (d) e/h

SECTION B (Short Answer - 3 Marks)

Q4. Define a wavefront. Using **Huygens' Principle**, verify the laws of Refraction (Snell's Law) at a plane surface.

Mandatory Derivation: 100% Frequency

Q5. Using Bohr's postulates for the hydrogen atom, derive the expression for the **radius of the n th orbit**. Show that $r \propto n^2$.

SECTION C (Case Study - 4 Marks)

Case Study: Lens Maker's Formula. High value lenses in astronomical telescopes require specific curvature. A double convex lens has $r_1=10\text{cm}$, $r_2=15\text{cm}$ and $n=1.5$.

(i) Calculate its focal length. (ii) If the lens is immersed in water ($n_w=1.33$), calculate the new focal length and describe the change.

SECTION D (Long Answer - 5 Marks)

Q6. Draw a labelled ray diagram of an **Astronomical Telescope** in normal adjustment. Derive the expression for its magnifying power.

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Draw a labelled ray diagram of a **Compound Microscope**. Derive magnification when image is formed at LDDV.

Q7. Explain the principle and working of a **Moving Coil Galvanometer**. Explain the function of the radial magnetic field.

--- OR ---

Explain **Full Wave Rectification** using p-n junction diodes. Draw the circuit and input/output waveforms.

Q8. Derive the expression for **Capacitance of a Parallel Plate Capacitor** when a dielectric slab is inserted between the plates.

--- OR ---

Derive the expression for **Electric Potential** at any point on the axis of an electric dipole.